When Reported Answers and Scored Actions Diverge

An Execution Audit of Public AutumnBench Change Detection

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Abstract. Environment-level queries require a coherent mapping from an agent's reported answer to its score. We audit this mapping in two publicly released AutumnBench Change Detection tasks using pinned native code. Across 136 controlled scenarios, submission grades an action-counter-derived index rather than the explicitly selected historical frame, although that frame is displayed correctly. At matched interface cost, different answers tie; repeating the same selection lowers the score. An index-only shadow evaluation changes all 128 explicit-selection scores. Two executed scripted reporting policies receive 0.5742 and 0.7100, while a separately labelled observation-based metric reverses their order. Paired original/changed replays also separate recorded guard time from first grid divergence. These findings concern the tested public implementation; they do not establish an error in published model comparisons.

1 The answer-to-score contract

WorldTest evaluates world-model learning through environment-level queries, instantiated in AutumnBench as masked frame prediction, planning and Change Detection (CD) [1]. Interpreting such scores requires the world history, explicit answer and evaluation target to remain aligned through the interface. The question here is whether the released CD pipeline preserves an agent's selected historical answer through submission and scoring.

Figure 1 separates semantic information from interface bookkeeping. The central execution finding is specific: selection displays frame k, but the following Submit assigns another index m before the reward reads it. The public code is executable; faithfulness of the answer-to-score mapping remains a separate property.

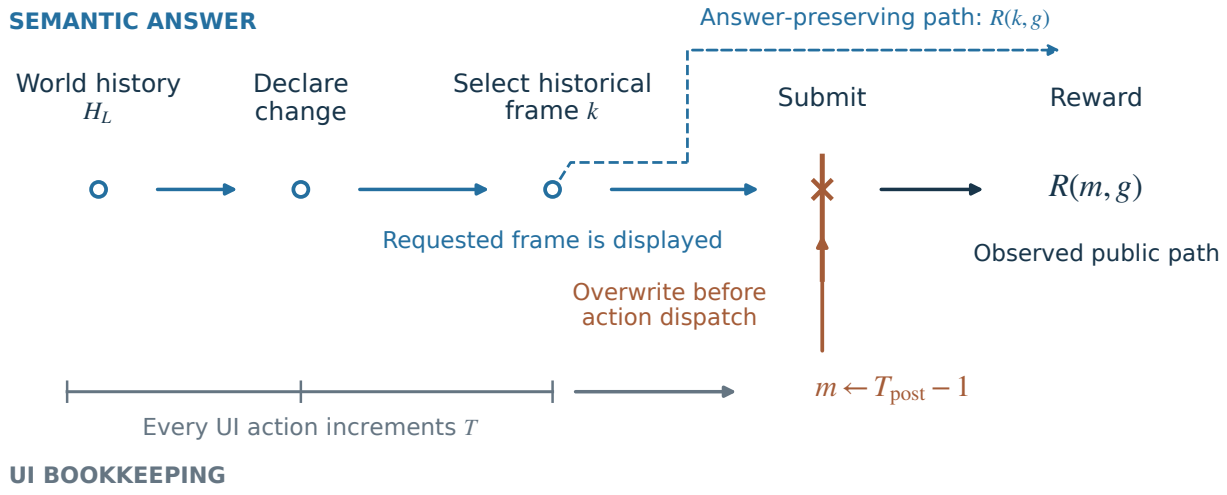


Figure 1. Where the answer diverges from the score. Blue tracks the historical answer; gray tracks the UI counter T. The orange assignment overwrites the selected index before action dispatch. The solid path is observed execution, with recorded guard time g. The dashed answer-preserving path is an interpretation of the published selection protocol, not confirmed author intent. Here R denotes the public reward function.

Scope boundary. The release contains 20 environments/60 tasks; the paper describes 43/129. This audit executes only CD in `27VWC` and `7WWW9`. No learned agent, published trajectory rescoring or full-benchmark reproduction is involved.

2 Protocol, execution path and controls

2.1 What is selected, and what is counted?

The published text interface describes declaration, historical-frame selection and confirmation [1, Appendix D.2 and Listing D.3]. Let L be the number of test-world actions, n the number of selection actions, k the explicit selected frame and T the interface action counter. Let g denote the *recorded trigger time*, not the Boolean guard predicate.

The pinned `CDSliderEnvironment.step()` increments T and assigns `curr_frame = time-1` before dispatch. Selection then sets k, but Submit performs the common assignment again before calling the reward [2]. For the executed sequences (initial $T = 0$, one declaration, n selections, one Submit),

$$T_{\text{post}} = L + n + 2, \qquad m = T_{\text{post}} - 1 = L + n + 1. \tag{1}$$

This is a source-derived rule tested against the saved inputs in all 136 factorial scenarios, including eight direct-submit controls with $n = 0$.

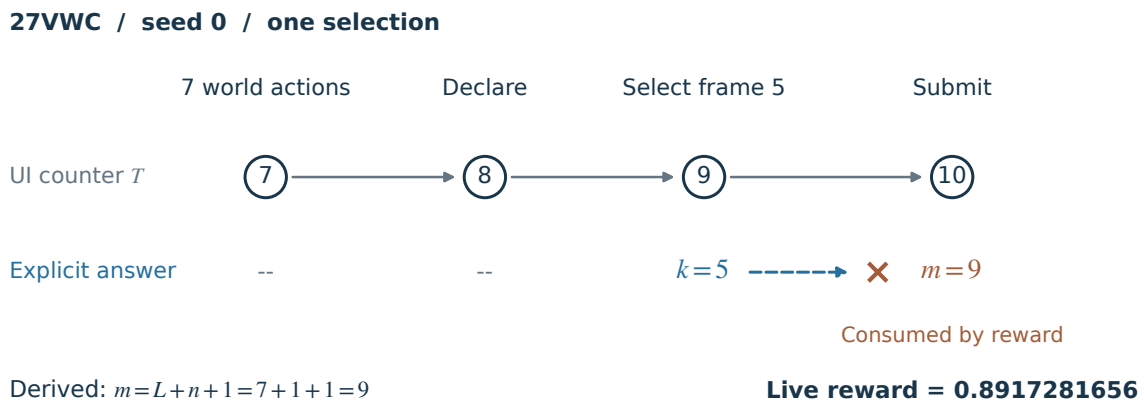


Figure 2. A recorded Submit transition. After six refill clicks and one noop, declaration and selection leave $T = 9$ and $k = 5$. Submit advances T to 10 and grades $m = 9$, returning 0.8917281655715406. The circles and index values are observed trace fields; Eq. 1 is the derived explanation. The stored history has indices 0–7; Submit returns feedback rather than displaying frame 9.

2.2 Real public execution, bounded design

The harness used the actual composite servicer and official generated protobuf requests in process:

$$\text{Initialize} \to \text{Reset} \to \text{initial observation} \to \text{go-to-test} \to \text{changed observation}$$
$$\to \text{world actions} \to \text{declaration} \to \text{selection} \to \text{Submit}.$$

All actions were checked against advertised spaces. Requests, returns and true scorer inputs were saved before assertions. The initial changed observation has $T = 0$ and one stored frame. Source/native code were unchanged; reward logging delegated to the original scorer.

Table 1. Executed design. Each environment uses one fixed world prefix, seeds 0 and 1729, and two repetitions. These are dependent diagnostic scenarios.

Stage	Selection design	Scenarios
27VWC primary	Four frame rules × browsing counts $\{1, 2, 4\}$; four no-selection controls	52
7WWW9 conditional	Same design; gated on primary effect and control agreement	52
7WWW9 exploratory	Frames $\{4, 5, 6, 7\}$ × browsing counts $\{1, 4\}$	32
Total	Excludes three path controls and eight paired native replays	136

Native, metered-direct and composite controls agree on the representative 27VWC trace. The first 104 scenarios were preregistered; the exploratory 32-scenario plan was frozen after their results and before its own run.

3 Scoring consequences and a controlled counterexample

3.1 Official execution and shadow evaluation are different objects

Observed. At fixed browsing count, all four selected answers receive identical live rewards in 32/32 groups. Holding the answer fixed, one versus four selections lowers the live reward in 48/48 groups. All 128 selected scenarios display the requested historical grid. Exact repetitions agree in 68/68 record pairs; these counts are not independent benchmark samples.

To isolate the consumed-index effect, the harness first saved the complete official Submit result, then shallow-copied the terminated Python UI object, changed only `curr_frame` to k and invoked the unchanged unbound public reward function. All other recorded scorer inputs remained equal. The official instance was unchanged and zero extra native calls were made. This *index-only shadow* changes 128/128 explicit-selection scores while retaining the public guard and raw reward scale. It is not a live benchmark run or a validated full repair.

3.2 Two executed reporting policies

The support-aware extension compares two scripts on the same 7WWW9 world history. Policy C selects frame 6 four times; E selects frame 4 once. Both then Submit. Table 2 preserves the full saved values.

Table 2. A scripted score-order reversal. $R_{\text{pub}}(m; g)$ is the live public score; $R_{\text{pub}}(k; g)$ is the index-only shadow. $S_{\text{obs}}(k; d)$ is an independent metric defined in Section 4. Here $g = 7$ and $d = 6$.

Policy	k	n	m	Live $R_{\text{pub}}(m; g)$	Shadow	S_{obs}
C	6	4	17	0.574152245828063	1	1
E	4	1	14	0.710035689635236	−1	0

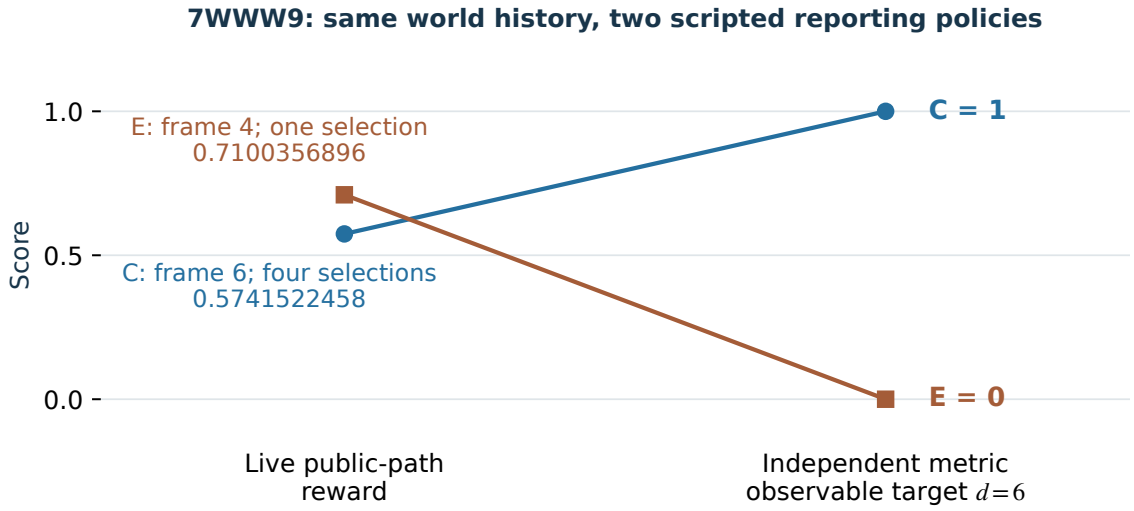


Figure 3. The score orders these reporting scripts differently. The live public score favors E; the independent observable-target metric favors C. Lines connect the same scripts under two scoring definitions, not learning curves or model-performance averages. The visible-target interpretation is conditional on Section 4; the live rewards are observed. All four seed/repetition blocks repeat the example exactly, with no inferential error bars.

Inferred. Under the deterministic observable-target interpretation, frame 6 is correct and frame 4 is too early. Their official ordering is reversed relative to this answer-based metric. Unequal browsing is the manipulated variable, not an efficiency-matched model comparison. The fixed-cost four-answer controls independently show official ties even when answers differ.

The eight no-selection controls retain null shadow scores: an explicit answer is not invented. In the factorial, early raw −1 values occur in the shadow calls, not the live Submit returns. No learned-agent ordering or published model ranking was evaluated.

4 A separate target-semantics question

4.1 Recorded trigger versus visible incompatibility

The paper defines change time t^* through the first realized observation prefix outside the original environment's support, and requires its guard to coincide with that time [1, Appendix C.2]. The public evaluator instead records a supplied condition when polled. Equality of that recorded time and a visible-history boundary is a substantive assumption.

Observed. Eight paired replays executed the exact public original and supplied changed programs under identical actions, with two seeds, two repetitions and a 128-step horizon. In `27VWC`, six refill clicks are followed by noop; `7WWW9` uses the fixed 12-arrow prefix before noop. Every official world prefix in the 136 scenarios matches its corresponding native changed replay.

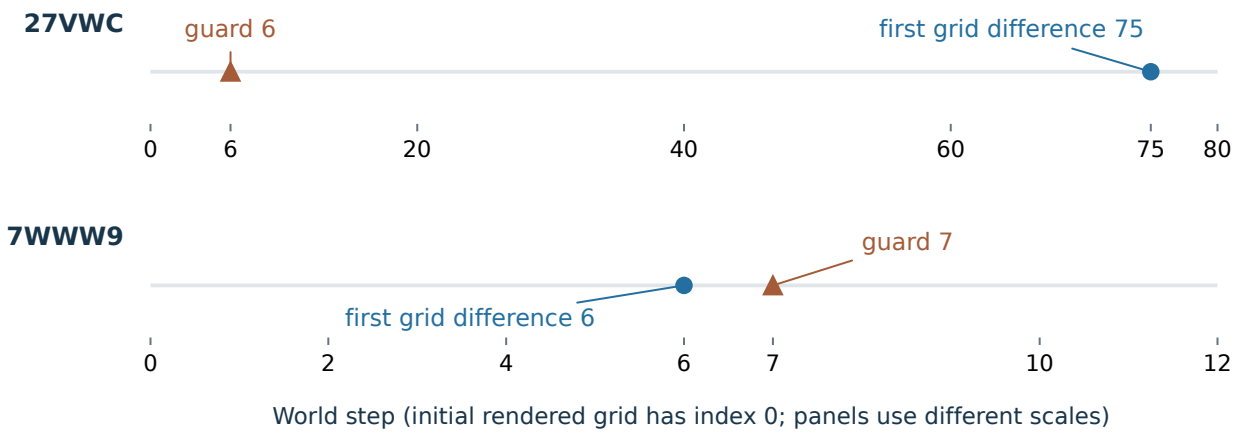


Figure 4. Two quantities that do not coincide in these replays. Triangles mark recorded guard time; circles mark first rendered-grid divergence from the original program. The pairs are $(6, 75)$ for 27VWC and $(7, 6)$ for 7WWW9. Display ranges differ (0–80 and 0–12); all saved replays extend to 128 world steps. Grid divergence is observed. Its support interpretation requires the deterministic, fixed-initialization assumptions below.

Trajectories repeat exactly across seeds and repetitions. The inspected originals have fixed initialization and no stochastic calls on these paths. Source inspection supports a latent gas/refill change before the coarse visual threshold in 27VWC, and next-action guard polling after the visible step-6 magnet movement in 7WWW9.

Inferred. Under those deterministic released-program assumptions, fixed initialization and actions give a point-mass original grid trajectory; its first different grid is thus an impossible original prefix. This interpretation is relative to the pinned interpreter and rendering. A same-seed difference alone does not prove support departure for stochastic dynamics, other initial-state distributions or other modalities. Recorded g and divergence d remain separate outside these assumptions.

4.2 What the independent metric changes

For interpretation only, the saved analysis uses the paper's piecewise scoring formula with d as target: score 0 for $k < d - 1$, 1 for $k \in \{d - 1, d\}$, and the published late branch otherwise. This is $S_{\text{obs}}(k; d)$ in Table 2. It changes target and, for early answers, raw scale; it must not be confused with the index-only shadow $R_{\text{pub}}(k; g)$.

The distinction matters empirically. Frame 5 in 7WWW9 is full-credit under $d = 6$ but receives -1 in the guard-7 shadow. Thus the initial guard-based comparison did not make frame 5 a clearly wrong observable-time answer. The exploratory extension used frame 4 instead and retained the original results. In 27VWC all 48 explicit selections occur before frame 74; their independent score is 0 despite positive live rewards.

Preserving the answer alone does not settle target identity or raw scale. Latent activation, declaration latency and observable incompatibility are distinct possible targets; choosing one requires a stated contract. Author intent and published-run provenance remain unconfirmed.

5 Implications, limits and focused author questions

5.1 Measurement validity, rather than a new learner

The executed public path mixes explicit answers with UI bookkeeping. This is relevant to the interpretation of an environment-level CD score even when the native world history is faithfully reproduced. The causal index comparison and fixed-cost answer controls make the concern more than a static representation difference.

Software testing already studies metamorphic relations and difficult test oracles [3]. Recent oracle-integrity work explicitly audits mutable scoring state and uses known-answer controls [4]. Reliable evaluation also requires attention to experimental dependence and design [5]. This note contributes a bounded execution case study and its target qualifications, not a new general testing principle or validated repair framework.

Proposed; untested. An evaluator could be checked for answer preservation, nuisance invariance, answer sensitivity and target consistency. Invariance is appropriate only when the task does not explicitly charge for the changed UI operation. A future study would need legitimate non-invariant controls, independently specified targets and held-out implementations; stochastic support reasoning would require more than paired seeds. No transfer rate or general detection guarantee has been established here.

5.2 What these results do not establish

There are only two executed public tasks and one frozen world prefix per environment. Seeds and repeated deterministic outcomes are dependent. The conditional second environment was gated on a primary effect; the support-aware extension used earlier observations. No population prevalence, confidence interval or learned-agent ability effect follows from these counts.

The exact scoring revision, graphical interface, trajectories and postprocessing used for the paper's experiments are not identified. Static inspection of the released example agent and controller does not bridge that gap. The paper's reported human/model gap and model rankings were not re-evaluated. The tested raw scorer uses -1 for early answers, whereas the published formula uses 0; any later mapping remains unverified.

The native/meter/direct-versus-composite agreement is measured on one representative 27VWC trace. The reward observer delegates to the original method, but no separate observer-off trial was run. Socket gRPC transport, mid-episode reset, no-change cases and stochastic support estimation are outside this execution audit.

5.3 Questions that would resolve the remaining contract uncertainty

1. Which exact revision, interface and reward postprocessing generated the published CD scores?
2. Should grading consume the explicitly selected historical index, declaration time or submission time? Are post-declaration browsing actions intended to incur a score penalty?
3. For these two frozen traces, how should recorded trigger time relate to latent rule activation and the first impossible original observation prefix?
4. Where, if anywhere, is raw early reward -1 mapped to the published scoring scale?

Takeaway. The public path demonstrably changes scores through submission bookkeeping. A useful next step is to resolve its answer/target contract and document a separately labelled evaluation variant if needed. That supports a focused technical discussion; it does not justify correcting published results or resuming the stopped full intervention study.

A Provenance and reproducibility

A.1 Exact public versions

Component	Frozen identity
MARAProtocol [2]	`d0d4e9251151778415cc5195c5dfb8296a5ded24`
Autumn.cpp [6]	`0929d175c3d296de091ac4b998d09b9b55a38f67`
Public assets [7]	Zenodo record 19498269; 20 environments/60 tasks

The tested `concrete_envs.py` SHA-256 is

`3cec4a2ce409867c756e0e0a8960b524ac5a251804303f33f03b09f1c795842d`.

Public checks on 24 October matched both pins and scorer bytes; published-run provenance remains unknown.

The companion evidence package is

`autumnbench_research_continuation_20261024`.

It retains frozen plans, raw logs, scorer inputs, hashes and an offline verifier. It is privately archived; no public evidence repository is asserted.

Evidence paths. Stage records are in `execution_evidence/results/`; continuous actions and native calls are in `execution_evidence/raw/`. Design files are `PREREGISTRATION.json` and `SUPPORT_EXTENSION_PLAN.json`. Derived checks are in `ANALYSIS_SUMMARY.json`; the offline entry point is `scripts/verify_saved_evidence.py`.

The 139 CD submissions include 136 factorial scenarios and three controls. All 1,978 action requests have saved returns, with zero action exceptions. There are 21,513 metered native calls (one unmetered control excluded). Runner wall time was 13.1634 seconds; CPU core-seconds were not measured. This note adds no experiment.

Correction and preservation. Pinned `InteractiveEnvironment.__init__` assigns `self.prog` at line 47; the earlier compatibility report's contrary claim was incorrect. No constructor adapter was needed. Both old failed attempts and the two-page note remain unchanged; their missing historical CD reward is not imputed from new traces.

References

[1] Archana Warrier, Dat Nguyen, Michelangelo Naim, Moksh Jain, Yichao Liang, Karen Schroeder, Cambridge Yang, Joshua B. Tenenbaum, Sebastian Josef Vollmer, Kevin Ellis, and Zenna Tavares. Benchmarking world-model learning with environment-level queries. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 133215–133246, 2026. https://proceedings.mlr.press/v306/warrier26a.html.

[2] Basis Research. MARAProtocol: Official public implementation. https://github.com/BasisResearch/MARAProtocol. Revision pinned in Appendix A; inspected 24 October 2026.

[3] Xiaoyuan Xie, Joshua W. K. Ho, Christian Murphy, Gail Kaiser, Baowen Xu, and Tsong Yueh Chen. Testing and validating machine learning classifiers by metamorphic testing. *Journal of Systems and Software*, 84(4):544–558, 2011. https://doi.org/10.1016/j.jss.2010.11.920.

[4] Junjie Chen. When agent evaluation becomes unreliable: An empirical study of oracle failures and configuration-dependent measurement in an LLM agent benchmark. arXiv:2610.16067v2, 2026. Preprint, revised 21 October 2026. https://arxiv.org/abs/2610.16067v2.

[5] Andrew Patterson, Samuel Neumann, Martha White, and Adam White. Empirical design in reinforcement learning. *Journal of Machine Learning Research*, 25(318):1–63, 2024. https://jmlr.org/papers/v25/23-0183.html.

[6] Basis Research. Autumn.cpp: Native autumn interpreter. https://github.com/BasisResearch/Autumn.cpp. Revision pinned in Appendix A; inspected 24 October 2026.

[7] Archana Warrier et al. AutumnBench public benchmark assets. Zenodo, record 19498269. https://zenodo.org/records/19498269, 2026. Frozen public subset: 20 environments and 60 tasks; CC BY 4.0.